

1(a)(i)	any two from: 1 idea of stopping heat (from lamp) reaching plant ; 2 (so) <u>temperature</u> does not change ; 3 (because) <u>temperature</u> affects, the rate of photosynthesis / enzymes (involved in photosynthesis) ;	2
1(a)(ii)	to provide carbon dioxide ;	1
1(a)(iii)	<u>oxygen</u> ;	1
1(b)(i)	as light intensity increases the rate of photosynthesis increases / ora ; data quote (two values of rate of movement of air bubble plus two values of distance of lamp from pond, plus units) ;	2
1(b)(ii)	light intensity is no longer a limiting factor ; temperature / carbon dioxide (concentration), could be the limiting factor ;	2

Question	Answer	Marks
1(c)	<i>any five from:</i> 1 ref. to (only) photosystem 1 / P700 ; 2 light <u>energy</u> absorbed by, chlorophyll <u>a</u> / primary pigment / reaction centre ; 3 electrons, excited / move to higher energy level ; 4 (electron) emitted by, chlorophyll <u>a</u> / primary pigment / reaction centre ; 5 (to) electron, carrier / acceptor ; 6 passes along, <u>electron transport chain</u> / <u>ETC</u> ; 7 chemiosmosis / description ; 8 (leading to) ATP synthesis ; 9 electron returns to, photosystem 1 / P700 / chlorophyll <u>a</u> / primary pigment / reaction centre ;	5